

Matter Is Made of Particles

Break chalk, dissolve sugar — both show matter is made of huge numbers of tiny particles with spaces between them.

The building blocks of matter

If we keep breaking chalk — snapping, then grinding — we reach the smallest units it is made of. These tiny basic units are the **constituent particles**. One piece of chalk contains a huge number of them.

Grinding chalk is a physical change: the chalk stays chalk — only the size of each speck reduces.

Dissolving sugar

Stir sugar into water and it disappears, yet the water tastes sweet everywhere. The sugar broke into its **constituent particles**, which are too small to see (even under an ordinary microscope) and slipped into the gaps between the water particles.

- The gaps between particles are called **interparticle spaces**.
- The water level barely rises because sugar particles fit into these spaces.

Scientific heritage: the ancient Indian philosopher **Acharya Kanad** proposed that matter is made of tiny indivisible particles called **Parmanu** (in his *Vaisheshika Sutras*).

Where students lose marks

Trap 1 — Dissolving sugar is a physical change. The sugar isn't destroyed; it splits into particles spread through the water (you can taste it).

Trap 2 — "Can't see it" doesn't mean "gone". Particles too small to see are still there — sense them by taste, smell, etc.

Trap 3 — Interparticle spaces are the gaps, not the particles. Sugar fits *into* the spaces between water particles.

Model answers — write them like this

Example A. Why does the water level hardly rise when sugar dissolves in it?

Step 1: Sugar breaks into very tiny constituent particles.

Step 2: These fit into the interparticle spaces between water particles.

∴ The sugar occupies existing gaps, so the level barely changes.

Example B. Is grinding chalk into powder a physical or chemical change? Why?

Step 1: Grinding only reduces the size of each speck.

Step 2: No new substance forms — it is still chalk.

∴ It is a physical change.

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